

Taylor & Francis
Taylor & Francis Group

Production Planning & Control
The Management of Operations

ISSN: 0953-7287 (Print) 1366-5871 (Online) Journal homepage: www.tandfonline.com/journals/tppc20

Designing supply chains with sustainability considerations

Amin Chaabane, Amar Ramudhin & Marc Paquet

To cite this article: Amin Chaabane, Amar Ramudhin & Marc Paquet (2011) Designing supply chains with sustainability considerations, Production Planning & Control, 22:8, 727-741, DOI: 10.1080/09537287.2010.543554

To link to this article: https://doi.org/10.1080/09537287.2010.543554

Published online: 09 Feb 2011.

Submit your article to this journal

Article views: 3763

View related articles

Citing articles: 25 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=tppc20

Production Planning & Control
Vol. 22, No. 8, December 2011, 727–741

Taylor & Francis
Taylor & Francis Group

Designing supply chains with sustainability considerations

Amin Chaabane*, Amar Ramudhin and Marc Paquet

Department of Automated Manufacturing Engineering, École de Technologie Supérieure, 1100, Notre Dame Street Ouest, H3C 1K3, Montreal, QC, Canada

(*Received 13 November 2010; final version received 18 November 2010*)

Increasing social awareness together with the new regulations for carbon and waste management are forcing enterprises to reconsider their supply chains with respect to economic, social and environmental objectives. Furthermore, cap and trade legislation for greenhouse gas emissions introduces a new level of complexity. This article presents a comprehensive methodology to address sustainable supply chain design problems where carbon emissions and total logistics costs, including suppliers and sub-contractors selection, technology acquisition and the choice of transportation modes, are considered in the design phase. The proposed methodology provides decision makers with a multi-objective mixed-integer linear programming model to determine the trade-off between economic and environmental considerations. This methodology is illustrated through the study of a Canadian firm operating in the steel industry which is facing a new legislation that caps carbon emissions. The results show how emission trading market can be used to reduce the carbon dioxide abatement cost.

Keywords: sustainable supply chain; supply chain design; greenhouse gases; carbon trading; mixed integer programming; goal programming

1. Introduction

Sulphur dioxide caps for electric utilities in the United States, regulatory carbon dioxide caps for companies across the European Union and domestic regulatory framework for greenhouse gas (GHG) emissions reduction in Canada and Australia are only a few of the numerous regulations on air emissions that exist today. Corporations are realising that sustainability policies are bottom-line issues. Aberdeen Group argues through a survey of 300 firms, worldwide, that corporate social responsibility (CSR) and sustainable supply chain management (SSCM) are on the top of the 'green agenda' (Nari *et al.* 2008). The survey reports that 50% of the firms are planning to redesign their supply chain to be more sustainable and that almost 80% of them have to be in compliance with new environmental regulations.

As a consequence, corporations face new realities and need to evaluate new options and mechanisms to meet their legal obligations. Ideally, they should reduce carbon emissions through sustainable actions such as the implementation of energy efficiency measures, the deployment of carbon capture and storage systems or investing in other emission reduction strategies and green technologies. Alternatively, companies can earn carbon credits through other compliance mechanisms such as the contribution to climate change technology fund or through emission trading systems (ETS) and carbon trading (Peace and Juliani 2009). Carbon trading markets were first introduced under the Kyoto Protocol, known as the United Nations framework convention on climate change (UNFCCC), as a worldwide effort to reduce GHG emissions. Thus, with the emergence of a price on GHG emissions, what previously could be emitted with no economic consequence can now be bought or sold with those making the biggest reduction in GHG standing to gain the highest monetary value.

Emission trading scheme are based on two concepts, allowances and the cap-and-trade system. An allowance represents the right to emit a given quantity of GHG during a given period. Allowances are established by the government through environmental regulations and they are denominated at 1-tonne of CO_2 equivalent, expressed as tCO_2e. A company holding one allowance for a specified period may emit one tCO_2e during this period. A cap is a limit on GHC emission. Companies that reduce GHG emissions below the cap would be allocated tradable carbon credits. Those corporations exceeding the cap need to acquire an equivalent amount of allowances (from tradable carbon credits) to meet the regulatory

*Corresponding author. Email: amin.chaabane@etsmtl.ca

ISSN 0953–7287 print/ISSN 1366–5871 online
© 2011 Taylor & Francis
http://dx.doi.org/10.1080/09537287.2010.543554
http://www.tandfonline.com

obligation or stand to pay a penalty. There are already a number of active carbon markets for GHG emissions such as the European Union Emission Trading Scheme (EU ETS) in Europe which is the largest multi-national GHG emissions trading scheme in the world, the New Zealand Emissions Trading Scheme (NZ ETS) in New Zealand, the Chicago Climate Exchange in United State (Johnson and Heinen 2004, Peace and Juliani 2009) and more recently the Montreal Climate Exchange in Canada (MCeX).

The motivation behind a 'cap-and-trade' emissions reduction system is extremely simple: it is a choice between 'make or buy'. Companies with a cap will comply by either reducing their emissions through changes in their supply chain (make GHG reduction) or by buying carbon credits from the market or from someone who has carbon credits. A company that has reduced its emissions below the cap gets a carbon credit equivalent to an amount reduced below the cap. The latter can then be traded in the carbon market (Labatt and White 2007). Choosing between make or buy is a complex decision because of the many options and places available, at all stages of the supply chain, for carbon reduction such as changes in product design, new suppliers, alternative manufacturing options, alternative transportation modes, the use of reverse logistics (recycle, reuse, etc.) and the price of the various options. A comprehensive decision-making methodology that integrates the various choices as well as the consideration of emission trading in order to achieve GHG reduction targets in a cost effective manner is necessary and would be very useful to supply chain managers for the design and evaluation of sustainable supply chains.

This article extends the basic methodology for designing carbon market sensitive and green supply chains that was introduced in Ramudhin *et al.* (2008). Section 2 introduces the problem and presents an overview of the literature in the area of sustainable supply chain design. The solution methodology is detailed in Section 3. Section 4 presents a multi-objective mixed integer model to support decision makers in the generation of different supply chain configurations and the evaluation of their performances with respect to economic and environmental constraints. Emphasis is given on the use of environmental data and the carbon trading systems. Section 5 details the solution methodology and the approach is illustrated through a case study of a well-known Canadian firm which operates in the steel industry and is facing new regulatory legislations that cap carbon emissions. Section 7 concludes this article and discusses future research directions.

2. Problem description and literature review

The government of Canada is committed to develop and implement a long-term plan for GHG emissions and air pollution reduction. Their regulatory framework establishes targets for the reduction of GHG emissions for different industrial sectors (Government of Canada 2008), such as electricity generation, oil and gas, pulp and paper, iron and steel, etc. Corporations touched by this regulation will have an immediate need to reduce GHG emissions. They would like to do this efficiently and in a cost-effective manner.

Several options are available for firms seeking compliance and/or to minimise the economic impact of the regulations. The firms can reduce their emissions through abatement actions. They can decide to contribute to a technology fund, which would then act as a means of promoting the development, deployment and diffusion of technologies that reduce emissions of GHG. They can also use emission trading mechanisms, including inter-firms trading, emissions reduction credits from non-regulated activities and certain credits from the Kyoto Protocol's Clean Development Mechanism. Finally, the use of a one-time recognition of early action between 1992 and 2006 to reduce GHG should also be considered (Government of Canada 2008) for those companies that are eligible. Thus, companies across the country are struggling to find the best strategies to maintain an efficient and sustainable supply chain under the new environmental regulations and the emerging carbon markets (Peace and Juliani 2009).

Sustainable Supply Chain Network Design (S-SCND) recognises that the long-term competitive advantage should be achieved through the alignment of economic, social and environmental goals (Frota Neto *et al.* 2008). The objective of economic sustainability is to minimise the total logistic cost or maximise the profit of different supply chain activities throughout all the product life cycle stages: purchasing, production, warehousing, distribution and recycling. Environmental supply chain sustainability means that permanent environmental damages should be avoided. Energy should be used efficiently; waste (liquid and solid) should be treated and air pollution should be reduced through the use of cleaner energies or other technologies. As for social sustainability, the objective is to improve the quality of life of the communities in which the supply chain operates through various initiatives such as funding special projects (school, hospital, etc), noise reduction and community services.

There are a number of approaches that strive to address different aspects of the sustainable supply chain management. Recent papers (Srivastava 2007,

Seuring and Muller 2008) present a review of several elements related to supply chain sustainability: green design (Pistikopoulos and Hugo 2005), inventory management (Ferretti *et al*. 2007), production planning and control for remanufacturing (Luo *et al*. 2001), green manufacturing (Ferretti *et al*. 2007), product recovery (Jayaraman *et al*. 1999), reverse logistics (Sheu *et al*. 2005, Chan 2007), waste management (Ferretti *et al*. 2007), energy use (Dotoli *et al*. 2005) and GHG emissions reduction (Ferretti *et al*. 2007). It is not surprising to see that mathematical modelling-based methodologies are the most commonly used models. Indeed, these models can be embedded in decision support systems to test the efficiency of various supply chain configurations and operating strategies. The variety of optimisation techniques used to tackle these problems include multi-criteria mathematical models, dynamic programming, non-linear programming and Markov chains. The literature suggests that sustainable supply chain practices require the integration of important decisions at different levels of planning (strategic, tactical, and operational) while considering the balance or trade-offs between conflicting key-performance indicators (Ferretti *et al*. 2007, Guillen-Gosalbez and Grossmann 2009). Multi-objective optimisation (MOO), a well-established area within the field of operational research (Cohon 1978), is particularly suited for this type of problems.

Examples of supply chain studies that directly incorporate environmental costs are limited at present in the literature but they are critical because of the rising cost of carbon emissions. Carbon emission trading has been steadily increasing in recent years. According to the World Bank's Carbon Finance Unit, 374 million metric tons of carbon dioxide equivalent (tCO_2e) were exchanged through various projects in 2005, a 240% increase relative to 2004 (110 m tCO_2e) which was itself a 41% increase relative to 2003 (78 m tCO_2e). In terms of dollars, the World Bank has estimated that the size of the carbon market as follows: 11 billion USD in 2005, 30 billion USD in 2006 and 64 billion USD in 2007 (Karan and Philippe 2008). As an evidence, the carbon price in the EU ETS peaked at €25 in 2008. The MCeX was launched in May 2008, and although the prices are very volatile right now, they are estimated to reach $100 by 2020 (Government of Canada 2008). The regulatory framework considers emissions trading as one of the important measures available for industries to face up to the issue of GHG reduction.

The main contribution of this work resides in a methodology that incorporates the 'cap-and-trade system' and regulatory environmental constraints with the traditional economic decisions in supply chain, including suppliers and sub-contractors selection, technology acquisition and transportation modes configuration. This approach uses a mixed integer linear programming (MILP) model that facilitates strategic decision-making and provides a better understanding of how the supply chain would react to various forms and combinations of environmental regulations and technological advances.

3. Solution methodology

Traditionally, supply chain network design methodologies attempted to establish the best supply chain configuration that maximises the long-term economic performance. The decisions cover strategic planning including product design, sourcing and subcontracting choices, technology selection and production strategies, storage mechanisms, transportation system configuration and the integration of reverse logistics activities. We refer to these decisions as 'internal strategic mechanisms' available to the supply chains managers in order to achieve the economic performance (Figure 1). The environmental impact of the supply chain was mostly ignored at the design phase leading to the plethora of environmental problems (climate change, global warming, etc.). In many countries, as a response to these problems, different regulations (e.g. Kyoto Protocol) have been introduced to monitor the control of GHG emissions of supply chain activities for different industrial sectors.

In this research, the methodology considers environmental regulations in the form of GHG emissions limits (caps). Thus, the supply chain performance is evaluated based not only on the economic performance but also on the environmental performance measured by GHG emissions. To be in compliance with the regulation, we allow a wider choice of options and operating strategies (internal strategic mechanisms). Moreover, carbon trading mechanisms are available for consideration and they are considered as the external mechanisms available to be in compliance with the regulation. The S-SCND problem is formulated as a multi-objective MILP and finds the best supply chain design that minimises the total logistics costs and GHG emissions with an interaction with carbon market place (Figure 1).

As illustrated in Figure 1, the methodology at its core is essentially a supply chain design approach that links carbon emissions to financial data by putting a price tag on them. It identifies various alternatives, both internal and external to the firm, in order to meet GHG reduction targets at a cost effective manner. Internal alternatives include changes in technology or supply chain strategies while the external mechanism considers buying or selling of carbon credits.

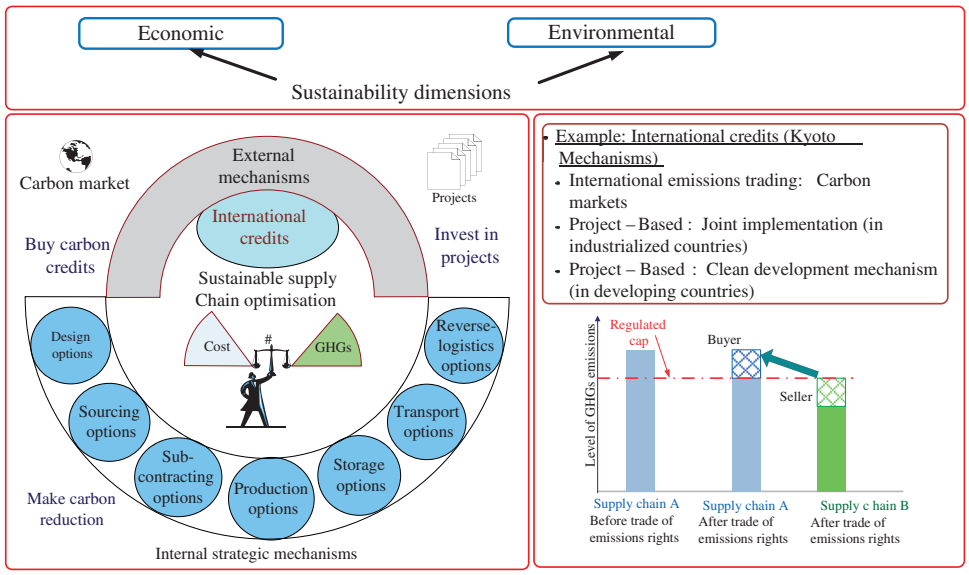


Figure 1. Sustainable supply chain design methodology.

Consider a supply chain where GHG emissions are more than its 'regulated Cap' (Figure 1). Two solutions are possible:

- if the cost of reducing one tCO_2e through internal strategic mechanisms is more than the carbon price in the market, it is better to buy carbon credits from the market;
- if the cost of reducing one tCO_2e through internal strategic mechanisms is less than the carbon price, it is better to implement these changes to achieve the target in term of GHG reduction.

Fundamental to the solution methodology is the use of an MILP to capture the interaction between the economic and the environmental dimensions. Two objective functions can be considered. The first one minimises the total logistic cost and the second minimises the total quantity of GHG emissions of the different supply chain activities. Sustainable supply chain network tries to find the ideal solution that minimises both cost and GHG simultaneously. This can be done through the use of a multi-objective technique such as 'goal programming' or 'ε-constraint' methods (Andersson 1999) to find a trade-off solution or the Pareto frontier curve.

4. Mathematical model

As shown in Figure 2, the supply chain considered is composed of different potential suppliers (V) from whom raw materials are purchased, a set of sub-contractors and plants (S) where products are manufactured and distributed to various customers zones (D) in different regions. Different technologies can be acquired to manufacture products and different transportation modes are used for product delivery between the nodes of the network (suppliers, sub-contractors, plants and customers). Although distribution and recycling centres stages are not considered in this study, it is very easy to add them in the model.

The two objective functions are considered. On the one hand, the objective of economic sustainability is to minimise the total logistics cost (F_1) of the supply chain, while on the other hand, the objective of environmental sustainability is to minimise the total emission quantity of GHG (F_2) calculated in units of tonnes of carbon dioxide equivalent (tCO_2e). The total logistic cost (F_1) is calculated as the summation of fixed costs (FC), variable costs (VC) and carbon credits component (CC):

$$F_1 = FC + VC + CC \tag{1}$$

Fixed costs are associated with the opening of facilities ($\sum_{i\in V\cup S}\lambda_i A_i$), technology acquisition ($\sum_{i\in S}\sum_{g\in G}\kappa_i^g W_i^g$) and assignment of raw material and production to the various sites ($\sum_{i\in V_p\cup S_p}\times \sum_{p\in R\cup M} a_{ip}Y_{ip}$) (see the Appendix for a full description of the variables and parameters):

$$FC=\sum_{i\in V\cup S}\lambda_i A_i+\sum_{i\in S}\sum_{g\in G}\kappa_i^g W_i^g+\sum_{i\in V_p\cup S_p}\sum_{p\in R\cup M} a_{ip}Y_{ip} \tag{2}$$

Variable costs are of four types: supply of raw materials ($\sum_{p\in R}\sum_{i\in V_p} b_{ip}X_{ip}$), production and

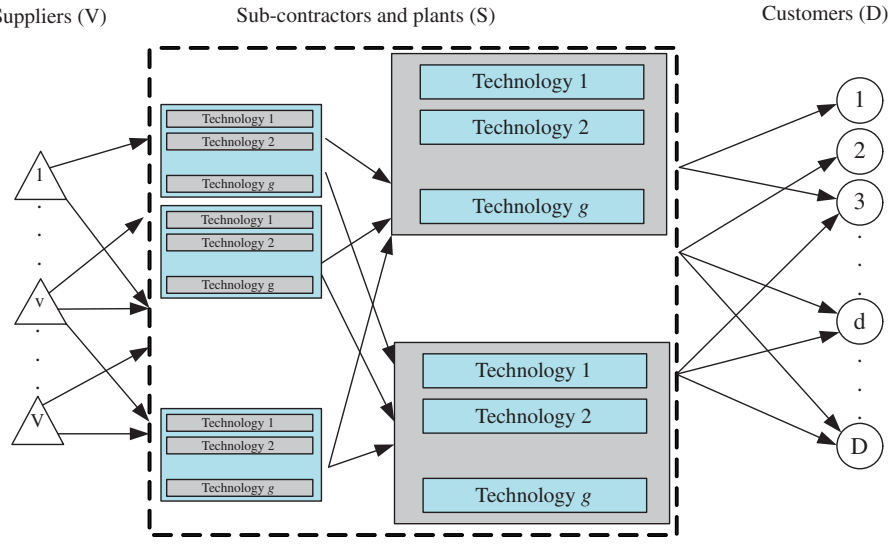


Figure 2. Supply chain network structure.

assembly of manufactured products ($\sum_{p\in R}\sum_{i\in S_p} \times \sum_{g\in G} c_{ip}^{g} Q_{ip}^{g}$), shipment costs ($\sum_{i\in S\cup V}\sum_{j\in S\cup D} \times \sum_{k\in K} l_{ij}^{k} U_{ij}^{k}$) and transportation costs ($\sum_{p\in M\cup R} \times \sum_{i\in S_p\cup V_p}\sum_{j\in SP_p^s\cup D}\sum_{k\in K} t_{ijp}^{k} F_{ijp}^{k}$).

$$VC = \sum_{p\in R}\sum_{i\in V_p} b_{ip}X_{ip} + \sum_{p\in R}\sum_{i\in S_p}\sum_{g\in G_p} c_{ip}^{g}Q_{ip}^{g} + \sum_{i\in S\cup V}\sum_{j\in S\cup D}\sum_{k\in K} l_{ij}^{k}U_{ij}^{k} + \sum_{p\in M\cup R}\sum_{i\in S_p\cup V_p}\sum_{j\in SP_p^s\cup D}\sum_{k\in K} t_{ijp}^{k}F_{ijp}^{k} \quad (3)$$

For the CC, we assume that the supply chain needs to be in compliance with a regulation that limits GHG emissions. Let L_{Emission} denote the limit (voluntary or mandated) on emissions known as the 'regulated cap' for the specific planning period. In this case, if the total carbon dioxide emissions, which is the summation of GHG emissions form transportation ($\sum_{p\in M\cup R} \times \sum_{i\in S_p\cup V_p}\sum_{j\in SP_p^s\cup D}\sum_{k\in K}\alpha^{k}\pi_p d(i,j)F_{ijp}^{k}$) and manufacturing activities ($\sum_{p\in M}\sum_{i\in S_p}\sum_{g\in G}\beta_{ip}^{g}\pi_p Q_{ip}^{g}$) are below the cap, the supply chain would be allowed tradable credits. Otherwise, there is a need to reduce emission through internal mechanisms or to buy an equivalent amount of carbon credits to meet regulatory obligation. Let ϕ denote the market price of an allowance under the carbon market. Thus, the carbon credit component is calculated as follows:

$$CC = \phi\left(\sum_{p\in M\cup R}\sum_{i\in S_p\cup V_p}\sum_{j\in SP_p^s\cup D}\sum_{k\in K}\alpha^{k}\pi_p d(i,j)F_{ijp}^{k} + \sum_{p\in M}\sum_{i\in S_p}\sum_{g\in G}\beta_{ip}^{g}\pi_p Q_{ip}^{g} - L_{\text{Emission}}\right) \quad (4)$$

A second objective is the evaluation of the supply chain from a purely environmental perspective. Thus, the objective of environmental sustainability is to minimise the total emissions quantity of GHG emissions (F_2) calculated in terms of tonnes of carbon dioxide equivalent (tCO_2e) is as follows:

$$\text{Min } F_2 = \sum_{p\in M\cup R}\sum_{i\in S_p\cup V_p}\sum_{j\in SP_p^s\cup D}\sum_{k\in K}\alpha^{k}\pi_p d(i,j)F_{ijp}^{k} + \sum_{p\in M}\sum_{i\in S_p}\sum_{g\in G}\beta_{ip}^{g}\pi_p Q_{ip}^{g} \quad (5)$$

The MILP supply chain network design model the typical design constraints such as supplier and subcontractors selection, demand and capacity constraints, manufacturing according to the specification of a bill of material (BOM), network structure constraints, emissions constraints and transportations constraints. The detailed model can be seen in the Appendix.

The proposed methodology for sustainable supply chain design considers the evaluation of the supply chain performance based on both objectives: (i) minimising the total logistic cost and (ii) minimising the environmental impact. In this particular case, the mathematical model can be formulated as following:

$$\text{Find: } \mathbf{X} = [x_1, x_2, \ldots, x_n]^T$$

$$\text{To minimize: } F(\mathbf{X}) = \begin{pmatrix} F_1(\mathbf{X}) = \text{Total Logistic Cost} \\ F_2(\mathbf{X}) = \text{GHG Emissions} \end{pmatrix}$$

$$\text{subject to: } \left\{ \begin{array}{l} g_j(\mathbf{X}) \le 0;\ j = 1, 2, \ldots, m \\ h_l(\mathbf{X}) = 0;\ l = 1, 2, \ldots, e \end{array} \right\}$$

$$\times \begin{cases} \text{Supplier and subcontractors selection constraints} \\ \text{Demand and capacity constraints} \\ \text{BOM Constraints} \\ \text{Network structure constraints} \\ \text{Emissions constraints} \\ \text{Transportation constraints} \\ \ldots\ldots \end{cases} \tag{6}$$

Here F is the utility function and $\mathbf{X}$ represents the vector of decision variables (continuous and binaries, respectively), belonging to the feasible region of equality $(h_l(X) = 0;\ l = 1, 2, \ldots, e)$ and inequality constraints $(g_j(X) \leq 0;\ j = 1, 2, \ldots, m)$, detailed in the Appendix.

Generally, a multi-objective optimisation problem can be handled in four different ways depending on when decision makers articulate their preference concerning the different objectives: never, before, during or after the actual optimisation procedure. In the first two approaches, the different objectives are aggregated to one overall objective function. Optimisation is then conducted with one optimal design as the result. The result is then strongly dependent on how the objectives were aggregated. Moreover, these methods do not use any preference information. Examples are the 'MinMax' formulation and global criterion method (Andersson 1999). The third approach is an iterative process where the decision-maker progressively articulates his preferences on the different objectives. They rely on progressive information about the decision-makers preferences simultaneously as they search through the solution space. In the fourth and final approach, optimisation is conducted without the decision-maker articulating any preferences among the objectives. The outcome of this optimisation is a set of Pareto optimal solutions which elucidate the trade-off between the objectives. The decision-maker then has to trade the objectives against each other in order to select the final design.

5. Solution method

In this article, we explore two solution procedures to solve the multi-objective optimisation problem for designing and evaluating sustainable supply chains. The first one is the 'ε-constraint' method. In this method (with posterior articulation of preference), one objective is selected for optimisation and the others are reformulated as constraints (Andersson 1999), i.e.:

$$\begin{pmatrix} \text{Find: } \mathbf{X} = [x_1, x_2, \ldots, x_n]^T \\ \text{To minimize: } F(\mathbf{X}) = \begin{pmatrix} F_1(\mathbf{X}) = \text{Total Logistic Cost} \\ F_2(\mathbf{X}) = \text{GHG Emissions} \end{pmatrix} \\ \text{s.t.: } \begin{Bmatrix} g_j(\mathbf{X}) \leq 0;\ j = 1, 2, \ldots, m \\ h_l(\mathbf{X}) = 0;\ l = 1, 2, \ldots, e \end{Bmatrix} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \text{Find: } \mathbf{X} = [x_1, x_2, \ldots, x_n]^T \\ \text{To minimize: } F(\mathbf{X}) = F_1(\mathbf{X}) \\ \text{s.t.: } \begin{Bmatrix} F_2(\mathbf{X}) \leq \varepsilon \\ g_j(\mathbf{X}) \leq 0;\ j = 1, 2, \ldots, m \\ h_l(\mathbf{X}) = 0;\ l = 1, 2, \ldots, e \end{Bmatrix} \end{pmatrix} \tag{7}$$

By progressively changing the constraint values, ε, which represent the limit on GHG emissions in this case, different points on the Pareto-front could be sampled. By calculating the extremes of the Pareto-front, the range of different objective functions could be calculated and constraint values selected accordingly. The second method is the goal programming (GP). The GP model could be placed in the third category. The algebraic formulation of GP is given as following:

$$\begin{pmatrix} \text{Find: } \mathbf{X} = [x_1, x_2, \ldots, x_n]^T \\ \text{To minimize: } \mathrm{F}(\mathbf{X}) = \begin{pmatrix} F_1(\mathbf{X}) = \text{Total Logistic Cost} \\ F_2(\mathbf{X}) = \text{GHG Emissions} \end{pmatrix} \\ \text{s.t.: } \begin{Bmatrix} g_j(\mathbf{X}) \leq 0;\ j = 1, 2, \ldots, m \\ h_l(\mathbf{X}) = 0;\ l = 1, 2, \ldots, e \end{Bmatrix} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \text{Find: } \mathbf{X} = [x_1, x_2, \ldots, x_n]^T \\ \text{To minimize: } Z = \sum_{i=1}^{2} (u_i n_i + v_i p_i) \\ \text{s.t.: } \begin{Bmatrix} F_i(\mathbf{X}) + n_i + p_i = F_i^*,\ i = 1, 2 \\ g_j(\mathbf{X}) \leq 0; j = 1, 2, \ldots, m \\ h_l(\mathbf{X}) = 0; l = 1, 2, \ldots, e \end{Bmatrix} \end{pmatrix} \tag{8}$$

where F_i^* is the target value for the objective function F_i which usually represents the minimum value obtained by considering this objective in the optimisation process; n_i and p_i represent the negative and positive deviations from this target value. The manager must analyse each one of the goals considered in the model in terms of whether over- or under-achievement of the goal is satisfactory where achievement implies that a goal has been reached. The terms u_i and v_i are the positive weights attached to these deviations in the achievement function Z. The weight factor of a given objective represents two different roles (Kettani *et al.* 2004). The first one is 'normalisation' that brings all deviations to a common unit of measurement.

The second is 'valorisation' reflecting the decision maker's preference structure. For instance, these weights take the value zero if the minimisation of the corresponding deviational variable is unimportant to decision makers. The 'ε-constraint' method first helps the decision maker to identify different possible solutions and the characteristic of each objective. Once he obtains, he can go through a decision process where he can articulate the preference structure and choose the trade-off solution that guarantee the different objectives.

5.1. *Optimisation methodology for sustainable supply chain design*

The methodology to design and evaluate sustainable supply chains is presented in Figure 3. In the first step, the problem is represented as a mathematical model (see Appendix for detailed model). Data for a particular instance of the supply chain are obtained from the enterprise information system and from other sources and stored in a database. Once the database is completed, the model is populated by a program developed in Microsoft Visual Basic 6.0. The program reads the data from the database and creates an LP file which is then solved by ILOG CPLEX®. At this step, decision makers are ready to begin the analysis and evaluation of the supply chain. If the importance of each objective is not completely known, a set of efficient solutions and the Pareto frontier curve using the 'ε-constraint' method can be generated. However, if they have preferences regarding the objective functions and their importance, then GP can be used with various weights to identify the best trade-off among the solutions.

6. Experimental evaluation

In this section, we consider the case of a firm that produces steel products with high levels of GHG emissions and that is subject to a regulation that caps carbon emissions. The different products are aggregated into one product family with two semi-finished products that are assembled from four parts sourced from various external suppliers. Indeed, it is commonly known that aggregated information about products is used especially when dealing with decisions at the strategic level. Three transportation modes are considered: rail, air and road to ship products between supply chain stages. In this study, emission is limited to carbon dioxide (CO_2) caused by production and transportation activities. Emission factors for the

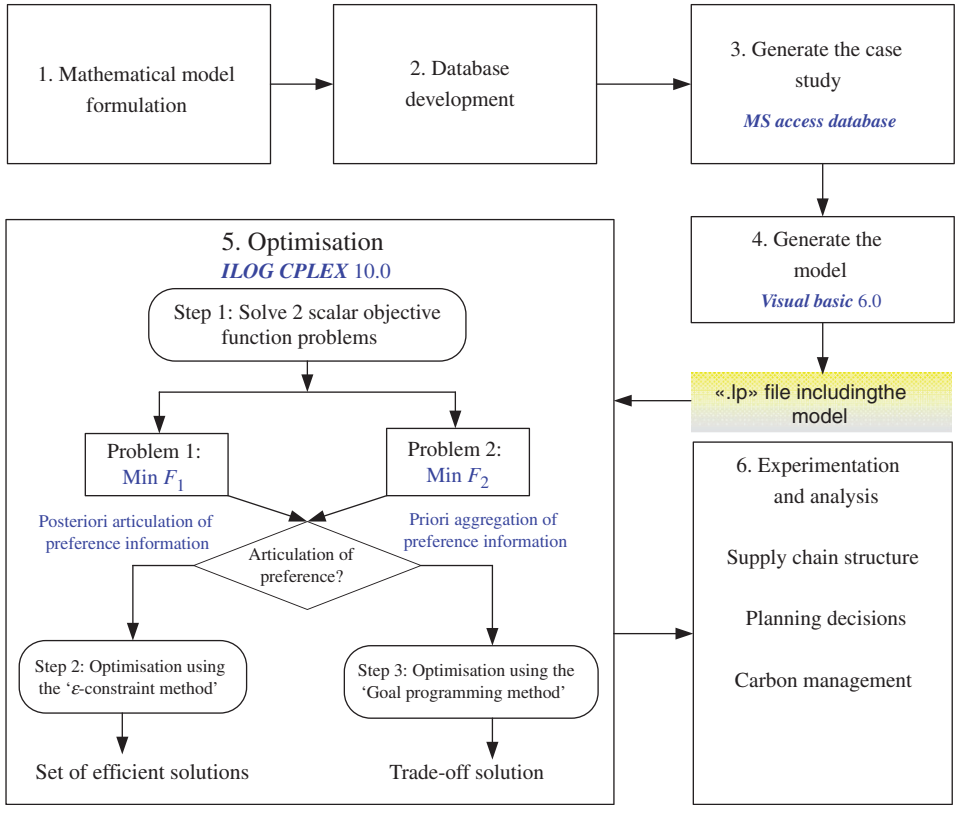


Figure 3. Optimisation methodology for sustainable supply chain design.

three transportation modes are detailed in Table 1. The emission factors (α^k) considered in this example are based on the recent accurate study published in Facanha and Horvath (2007). The manufactured products are primarily composed of steel materials. The emission factors (β_{ip}^g) for different technologies are obtained from the IPCC emission factor database [e.g. $\beta_{ip}^g = 1.6$ tons CO_2/ton steel product] (http://www.ipcc-nggip.iges.or.jp/EFDB/find_ef.php).

The model is first solved by CPLEX Interactive Optimizer 10.0 considering that only internal abatement mechanisms are available and there is no interaction with the carbon market. Two scenarios are analysed. In the first one, the model is solved considering only the objective function that minimises cost (F_1). The optimal cost is \$28,508,190. The total emission quantity relative to this solution is 80,191 tCO_2e. Next the model is solved for optimal levels of GHG emissions (F_2). The optimal GHG emissions quantity for the most environmental supply chain is 20,312 tCO_2e but at a total cost of \$44,935,790. Thus, if we suppose that the actual supply chain is optimised based on cost, a four-fold reduction in carbon emissions can be achieved at two times the cost of the current solution. Thus, the average abatement cost (AAC), which is the average cost of reducing one tone of CO_2 from the current situation, is equal to \$274 (Table 2).

In the second step, the integration of environmental regulation that caps GHG emissions as well as the interaction with the carbon trading market are considered. Under the regulatory framework for industrial GHG emissions (Government of Canada 2008), the limit of emission is fixed to 60,000 tCO_2 for the planning period, which represents a reduction of 25% of carbon emission when compared to scenario 1. The price of one tonne of CO_2 is assumed to be equal to \$15 (Government of Canada 2008). The model is solved again for the two scenarios. In the first case (scenario 1), we observe that the decision is the acquisition of carbon allowances from the carbon market and this represent an additional cost for the company (emission cost). However, if the purely environmental solution (scenario 2) is considered, a carbon credit for 39,688 tCO_2e (60,000 – 20,312) can be obtained for a price of \$595,315 (Table 3).

Table 1. Transportation modes and emissions factors (grams/ton-mile).

Modes	Type	Payload (tons)	CO_2 (grams/ton-mile)
Road	Class 8b	12.5	187
Rail	Intermodal rail	2093	40
Air	Boeing 747-400	70	1385

The average abatement cost is now equal to \$259. Figure 4 shows the cost breakdown of the extreme solutions (scenarios 1 and 2). As can be seen, production and transportation costs increase significantly because of the use of greener production technologies and more environmental transportation modes in scenario 2. However, the raw material cost is the same for both the scenarios. This means that supplier selection results remain the same for the two solutions.

To observe the sensitivity of the total logistics cost *versus* carbon emissions reduction, this following constraint is added to the model ('ε-constraint method) and solved for different values of upper bounds of total carbon emissions for the supply chain, and denoted UB_{Emission}.

$$\underbrace{\sum_{p\in M\cup R}\sum_{i\in S_p\cup V_p}\sum_{j\in S(Suc(P))\cup D}\sum_{k\in K}\alpha^k\pi_p d(i,j)F_{ijp}^k}_{\text{GHGs Emissions from transportation}} + \underbrace{\sum_{p\in M}\sum_{i\in S_p}\sum_{g\in G_p}\beta_{ip}^g\pi_p Q_{ip}^g}_{\text{GHGs Emissions from process}} \leq UB_{\text{Emission}} \quad (9)$$

Figure 5 shows that the total logistics cost decreases as the upper bound of CO_2e emissions (UB_{Emission}) increases, the model seeking less costly solution alternatives which have higher emission rates. From a managerial perspective, this means that those companies might have to look for new production or transportation alternatives and invest in environmental friendly technologies in order to reduce GHG emissions. However, the delivery lead time might increase in this case. Figure 5 also shows that

Table 2. Marginal abatement cost without carbon market integration.

Scenario	Scenario 1	Scenario 2	Comparison
Total cost (\$)	28,508,190	44,935,790	Cost increase by 158%
GHG emission (tCO_2e)	80,191	20,312	GHG reduced by a factor of 4
			AAC = \$274

Table 3. Marginal abatement cost with carbon market integration.

Scenario	Scenario 1	Scenario 2	Comparison
Total cost ($)	28,811,054	44,340,790	Cost increase by 154%
GHG emission (tCO_2e)	80,191	20,312	GHG reduced by a factor of 4
Emission cost	302,864	(595,315)	
			AAC = $259

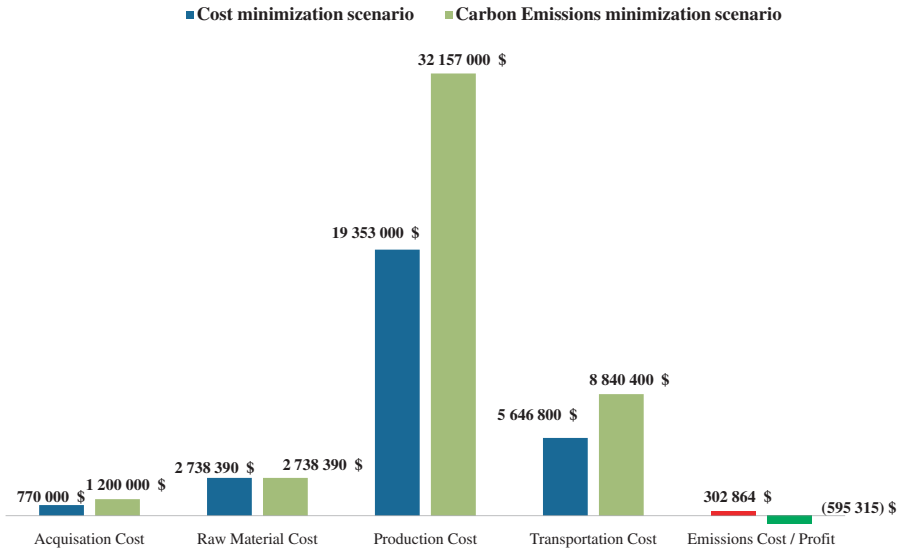


Figure 4. Cost analysis of extreme solutions (scenario 1 versus scenario 2).

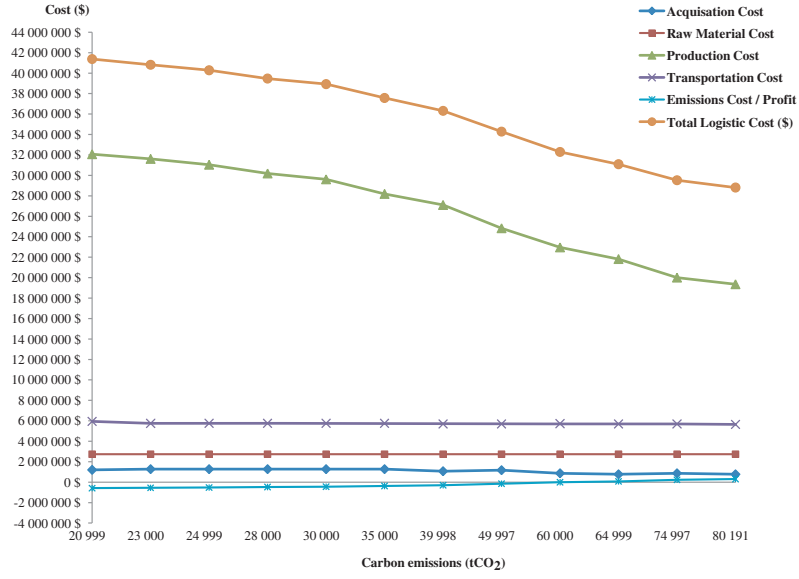


Figure 5. Logistic costs versus carbon emission.

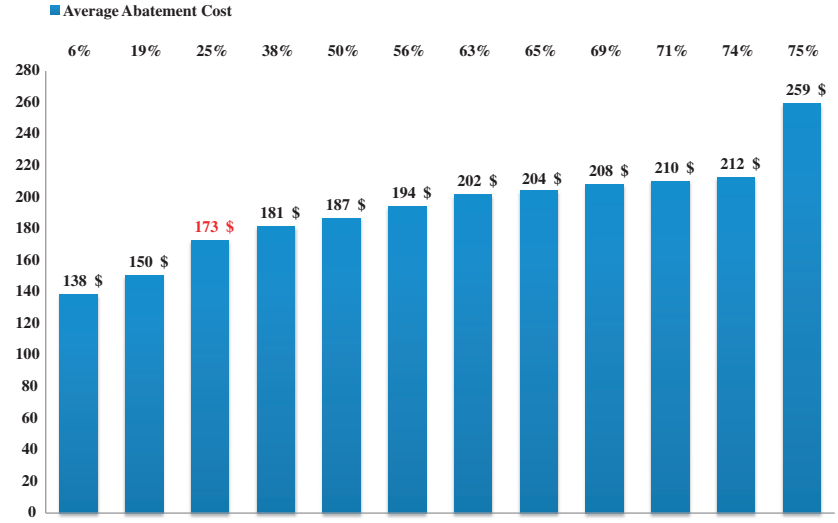


Figure 6. Average abatement cost *versus* carbon emissions reduction.

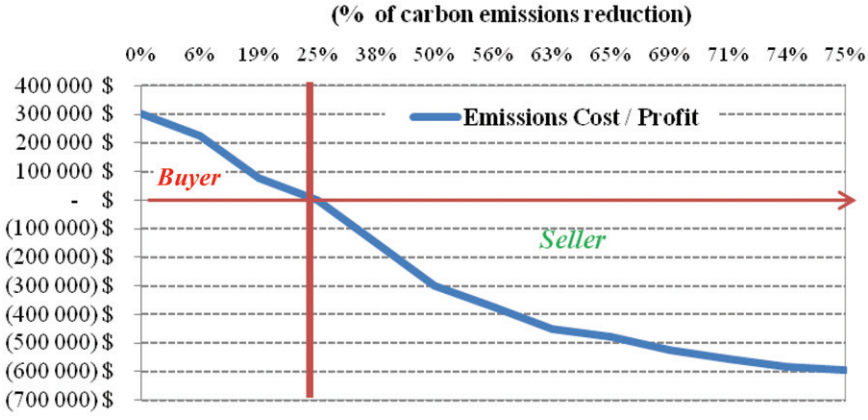


Figure 7. Emission cost/profit component.

the total logistics cost (F_1) and carbon emissions (F_2) are two conflicting objectives. Thus, the application of a multi-objective optimisation procedure could help to determine the best trade-off.

The interaction with the emission trading system helps the company to characterise exactly the average abatement cost of carbon emissions as a function of carbon emission reduction target (Figure 6). For example, if the objective is to reduce carbon emissions by 25%, then the average abatement cost is equal to $\$173/tCO_2$.

In addition, the decision in terms of buying or selling carbon credit from the carbon market is defined (Figure 7). In the case where the objective is to reduce carbon emissions less than 25%, the company should buy credits form the carbon market (buyer). However, if the target in term of GHG reduction is more than 25%, then the company might sell some carbon credit to reduce the impact of the environmental legislation on the economic objective.

The GP method was used for the previous example in order to find the trade-off between the total logistics cost and GHG emissions. Table 4 summarises the various solutions. The GP solution has an emission quantity of 49,312 tCO2e for an operational cost of $34,283,644. The total logistic cost for the GP solution is only 18% greater than the efficient scenario. This example demonstrates that by using a multi-objective approach, it is possible to look at the various solution trade-offs that reduces GHG emissions while maintaining logistics costs under control.

This case study demonstrates that the proposed methodology has the potential to be a tool for policy makers. The evaluation of the economic impact of regulation of the supply chain is identified with more

Table 4. GP solution.

Optimization scenarios	Total operational cost	tCO_2e
Scenario 1 – Cost minimisation	$F_1^* = \$28,811,054$	$F_2 = 80,191$
Scenario 2 – GHG emissions minimisation	$F_1 = \$44,340,790$	$F_2^* = 20,312$
Trade-offs scenario – GP	$F_1 = \$34,283,644$	$F_2 = 49,997$

accuracy and loopholes can be eliminated. For instance, the impact of non-homogenous carbon prices in varying geographies can have a negative impact on the social or environmental aspects of supply chain sustainability as they seek lower prices for carbon.

7. Conclusion

The main contribution of this article is the development of an integrated model for S-SCND that considers both internal strategic control mechanisms and leverages the opportunities offered by carbon trading markets (external strategic mechanisms). Using the model, supply chain managers are now able to determine the overall GHG footprint of supply chains operations. They can determine whether they qualify for carbon credits or must purchase credits on the carbon market place. That will help them decide on the best configuration strategy for their supply chain to be in compliance with regulations and achieve sustainability objectives.

Moreover, the proposed approach helps supply chain managers to evaluate the average abatement cost as a function of carbon emission reduction targets which is important to analyse within the context of the Kyoto Protocol as illustrated by the case study. From a managerial perspective, the solution methodology provides valuable insights into the design and evaluation of sustainable supply chain and guides decision-makers towards the adoption of the most cost-effective options as regulations become stronger.

The methodology presented here is general enough and may be applied to other supply chain studies to design sustainable supply chain and evaluate their performance in term of cost and carbon emissions. However, the mathematical model considers only a unique price at the time for carbon emissions which may not be realistic especially with the high volatility of carbon prices observed on last years at different regions and the expected change in the future. Thus, a multi-period model for S-SCND under the emission trading scheme where carbon prices are subject to possible changes with time and at different regions is important to add and subject to future development. Finally, the inclusion of other product life cycle stages such as distribution and reverse logistics activities should be integrated in the model to study more realistic supply chains of different sectors.

Notes on contributors



Amin Chaabane is currently a PhD student in the Department of Automated Manufacturing Engineering at École de Technologie Supérieure (ÉTS) in Montreal, Canada. He received his Bachelor degree in Industrial Engineering from École Nationale d'Ingénieurs de Tunis (ENIT), in 2002, and the Master of Science degree in Management Science (Logistics and Systems) from Conservatoire Nationale des Arts et Métiers (CNAM-Paris), in 2004. His research interests include linear and integer optimisation, multi-criteria and multi-objective optimisation, supply chain design, sustainable logistics, inventory, routing and scheduling.



Professor Amar Ramudhin teaches Operations and Logistics at École de Technologie Supérieure, Montreal, Canada, where he also heads the Supply Chain Research Laboratory. His main research interests are in the design of sustainable supply chains and the design and improvement of integrated business processes for complex systems with a keen interest in healthcare. He received his PhD from the Georgia Institute of Technology.



Marc Paquet is an Assistant Professor in the Department of Automated Manufacturing Engineering at École de technologie supérieure in Montreal. His research interests include manufacturing network design and optimisation, supply chain design, operation management and production planning and control. He holds PhD and MBA degrees from Université Laval in Quebec.

References

Andersson, J., 1999. *A survey of multiobjective optimization in engineering design*. Linköping, Sweden: Department of Mechanical Engineering, Linköping University.

Chan, H.K., 2007. A pro-active and collaborative approach to reverse logistics—a case study. *Production Planning & Control: The Management of Operations*, 18 (4), 350–360.

Cohon, J.L., 1978. *Multiobjective programming and planning*. New York: Academic Press.

Dotoli, M., *et al*., 2005. A multi-level approach for network design of integrated supply chains. *International Journal of Production Research*, 43, 4267–4287.

Facanha, C. and Horvath, A., 2007. Evaluation of life-cycle air emission factors of freight transportation. *Environmental Science & Technology*, 41, 7138–7144.

Ferretti, I., *et al*., 2007. Greening the aluminium supply chain. *International Journal of Production Economics*, 108, 236–245.

Frota Neto, J.Q., *et al*., 2008. Designing and evaluating sustainable logistics networks. *International Journal of Production Economics*, 111, 195–208.

Government of Canada 2008. *Turning the corner: regulatory framework for industrial greenhouse gas emissions*, Ottawa.

Guillen-Gosalbez, G. and Grossmann, I.E., 2009. Optimal design and planning of sustainable chemical supply chains under uncertainty. *American Institute of Chemical Engineers (AIChE) Journal*, 55, 99–121.

Jayaraman, V., Guide, V.D.R., and Srivastava, R., 1999. A closed-loop logistics model for remanufacturing. *Journal of the Operational Research Society*, 50, 497–508.

Johnson, E. and Heinen, R., 2004. Carbon trading: time for industry involvement. *Environment International*, 30, 279–288.

Karan, C. and Philippe, A., 2008. *State and trends of the carbon market 2008*. Washington, DC: The World Bank.

Kettani, O., Aouni, B., and Martel, J.-M., 2004. The double role of the weight factor in the goal programming model. *Computers & Operations Research*, 31, 1833–1845.

Labatt, S. and White, R.R., 2007. *Carbon finance: the financial implications of climate change*. Hoboken, NJ: John Wiley & Sons.

Luo, Y., Zhou, M., and Caudill, R.J., 2001. An integrated E-supply chain model for agile and environmentally conscious manufacturing. *IEEE/ASME Transactions on Mechatronics*, 6, 377–386.

Nari, V., Jhana, S., and Robert, S., 2008. *Supply chain network design: architecting a green future*. Boston: Aberdeen Group.

Peace, J. and Juliani, T., 2009. The coming carbon market and its impact on the American economy. *Policy and Society*, 27, 305–316.

Pistikopoulos, E.N. and Hugo, A., 2005. Environmentally conscious long-range planning and design of supply chain networks. *Journal of Cleaner Production*, 13, 1428–1448.

Ramudhin, A., *et al*., 2008. Carbon market sensitive green supply chain network design. *In: IEEE International Conference on Industrial Engineering and Engineering Management*, 8–11 December 2008, Singapore.

Seuring, S. and Muller, M., 2008. From a literature review to a conceptual framework for sustainable supply chain management. *Journal of Cleaner Production*, 16, 1699–1710.

Sheu, J.-B., Chou, Y.-H., and Hu, C.-C., 2005. An integrated logistics operational model for green-supply chain management. *Transportation Research Part E: Logistics and Transportation Review*, 41, 287–313.

Srivastava, S.K., 2007. Green supply-chain management: a state-of-the-art literature review. *International Journal of Management Reviews*, 9, 53–80.

Appendix

Sets and indices

In this study, the following sets and indices are used:

P	Set of all products
$R \subset P$	Set of raw materials
$M \subset P$	Set of manufactured products
$C \subset M$	Set of finished products
N	Set of all nodes
G	Set of manufacturing technologies
$D \subset N$	Set of customer zones
$S \subset N$	Set of all sub-contractors
$S_p \subset S$	Set of sub-contractors of product $p \in M$
$V \subset N$	Set of suppliers of raw materials
$V_p \subset V$	Set of suppliers of raw material $p \in R$
P_p^s	Set of immediate successors of product $p \in P/C$ in the BOM
SP_p^s	Set of sub-contractors for the set of immediate successors of product $p \in P/C$
M_i	Set of products that can be manufactured by sub-contractor $i \in S$
R_i	Set of raw materials that can be supplied by supplier $i \in V$
K	Set of all transportation modes $k \in K$

Parameters

The strategic mathematical model requires the following cost parameters:

λ_i	Fixed cost associated with the use of site $i \in S \cup V$
κ_i^g	Fixed cost associated with the acquisition of technology $g \in G$ at site $i \in S$
a_{ip}	The start-up cost associated with manufacturing product $p \in M$ at site $i \in S_p$
b_{ip}	Purchasing unit cost of raw material $p \in R$ at site $i \in V_p$
c_{ip}^g	Unit cost of producing product $p \in M$ at site $i \in S_p$ using technology $g \in G$
t_{ijp}^k	Transportation unit cost of product $p \in P$ from node $i \in V_p \cup S_p$ to node $j \in SP_p^s \cup D$ using transportation mode $k \in K$
l_{ij}^k	Cost of a single shipment between nodes $i \in V \cup S$ and $j \in S \cup D$ using transportation mode $k \in K$
ϕ	Price per metric ton of carbon dioxide equivalent (tCO_2e)

The following data are also needed:

- α^k GGHemissions factor per weight unit and per distance unit due to the use of transportation mode $k \in K$ per ton-mile
- β_{ip}^g Greenhouse gas emission factor (tones) per weight of produced quantity of product $p \in M$ using the technology $g \in G$ at node $i \in S_p$
- $L_{Emissions}$ Limit of emission fixed by government regulation
- $\theta_{pp'}$ Number of products $p \in P/C$ required to manufacture one unit of product $p' \in P_p^s$
- m_p Maximum number of sites that can be opened for product $p \in M \cup R$
- e_{ip} Capacity of node $i \in S_p$ for product $p \in R$ (supplier's capacity)
- f_i^g Available time at node $i \in S$ when using technology $g \in G$
- te_{ip}^g Processing time on product $p \in M$ at node $i \in S_p$ using technology $g \in G$
- d_{pd} Number of product $p \in C$ required by demand node $d \in D$
- ρ_i Lower bound (in %) on the aggregated capacity to be used if manufacturer or supplier $i \in S \cup V$ is chosen
- T_i Total time available at the assembly line of subcontractor $i \in S$
- τ_{ij} Maximum number of transportation modes that can be used between nodes $i \in V \cup S$ and $j \in S \cup D$
- κ^k Volume capacity of transportation mode $k \in K$
- ψ^k Weight capacity of transportation mode $k \in K$
- π_p Weight of product $p \in P$
- δ_p Volume of product $p \in P$
- $d(i,j)$ Distance between nodes $i \in V \cup S$ and $j \in S \cup D$

Decision variables

To find the optimal configuration of the network, the following decision variables are required:

- A_i Binary variable equals 1 if node $i \in V \cup S$ is open and operational for at least one product
- W_i^g Binary variable equals 1 if technology $g \in G$ is selected at node $i \in S$
- Y_{ip} Binary variable equals 1 if raw material $p \in R \cup M$ is assigned to node $i \in V_p \cup S_p$ and 0 otherwise
- X_{ip} Number of units of product $p \in R$ supplied by node $i \in V_p$
- Q_{ip}^g Number of units of product $p \in M$ manufactured by node $i \in S_p$ using technology $g \in G$
- F_{ijp}^k Number of units of product $p \in P$ shipped from node $i \in V_p \cup S_p$ to node $j \in SP_p^s \cup D$ using transportation mode $k \in K$
- U_{ij}^k Number of shipments between nodes $i \in V \cup S$ and $j \in S \cup D$ using transportation mode $k \in K$
- Z_{ij}^k Binary variable equals 1 if transportation mode $k \in K$ is used between nodes $i \in V \cup S$and $j \in S \cup D$ and 0 otherwise

Objective functions

In the following model, two objective functions are considered:

- Economic sustainability (F1): Minimise the total logistics cost of the supply chain considering fixed, variable, and emission costs.
- Environmental sustainability (F2): Minimise the total quantity of GHG emissions calculated in units of tons of carbon dioxide equivalent (tCO_2e).

Cost structure of the objective function F_1

Cost structure	Mathematical formulation
(1) Fixed cost for facilities	$\sum_{i \in V \cup S} \lambda_i A_i$
(2) Fixed cost for assignment products to sites	$\sum_{p \in R \cup M} \sum_{i \in V_p \cup S_p} a_{ip} Y_{ip}$
(3) Fixed cost for technology acquisition	$\sum_{i \in S} \sum_{g \in G} \kappa_i^g W_i^g$
(4) Fixed cost for transportation lanes	$\sum_{i \in V \cup S} \sum_{j \in S \cup D} \sum_{k \in K} l_{ij}^k U_{ij}^k$
(5) Raw materials cost	$\sum_{p \in R} \sum_{i \in V_p} b_{ip} X_{ip}$
(6) Manufacturing cost	$\sum_{p \in R} \sum_{i \in S_p} \sum_{g \in G} c_{ip}^g Q_{ip}^g$
(7) Transportation cost	$\sum_{p \in M \cup R} \sum_{i \in S_p \cup V_p} \sum_{j \in SP_p^s \cup D} \sum_{k \in K} l_{ijp}^k F_{ijp}^k$
(8) GHG Emissions cost/profit	$\phi \left(\underbrace{\sum_{p \in M \cup R} \sum_{i \in S_p \cup V_p} \sum_{j \in SP_p^s \cup D} \sum_{k \in K} \alpha^k \pi_p d(i,j) F_{ijp}^k}_{\text{GHGs Emissions from transportation}} + \underbrace{\sum_{p \in M} \sum_{i \in S_p} \sum_{g \in G} \beta_{ip}^g \pi_p Q_{ip}^g}_{\text{GHGs Emissions from production}} - \underbrace{L_{\text{Emission}}}_{\text{Cap}} \right)$

The GHG emissions cost/profit is calculated based on the credits compared to the limit of emissions $L_{Emission}$ fixed by regulations. Therefore, the objective function F_1 that represents the total operational cost to be minimised is

$$\text{Min } F_1 = (1) + (2) + (3) + (4) + (5) + (6) + (7) + (8) \quad \text{(A1)}$$

The objective function (F_2) is to minimise the total emissions quantity of GHG (tCO_2e) in order to evaluate the best potential reduction in term of GHG emissions.

$$\text{Min } F_2 = \underbrace{\sum_{p\in M\cup R}\sum_{i\in S_p\cup V_p}\sum_{j\in SP_p^s\cup D}\sum_{k\in K}\alpha^k\pi_p d(i,j)F_{ijp}^k}_{\text{GHGs Emissions from transportation}} + \underbrace{\sum_{p\in M}\sum_{i\in S_p}\sum_{g\in G}\beta_{ip}^g\pi_p Q_{ip}^g}_{\text{GHGs Emissions from manufacturing}} \quad \text{(A2)}$$

Constraints

For the MILP supply chain network design model, there are many constraints to be considered. These constraints are of many kinds including the balance constraints of all products, the capacity limit constraints, the minimum capacity occupation constraints and the demand satisfaction constraint. The BOM constraints are implicitly taken into account in the balance constraints. These elements are discussed below.

For each raw material and for each manufactured product, the number of operational sites should not exceed the maximum number allowed of suppliers and sub-contractors:

$$\sum_{i\in S_p\cup V_p} Y_{ip} \le m_p \quad (\forall p\in R\cup M) \quad \text{(A3)}$$

If a product (raw material) is assigned to a node (supplier), then the number of products supplied by this supplier must not exceed its capacity for this product:

$$X_{ip} - e_{ip}Y_{ip} \le 0 \quad (\forall p\in R,\ \forall i\in V_p) \quad \text{(A4)}$$

A product (semi-finished or final product) is manufactured in a node (sub-contractor) only if the product is assigned to this node:

$$\sum_{g\in G} Q_{ip}^g - MY_{ip} \le 0 \quad (\forall p\in M,\ \forall i\in S_p) \quad \text{(A5)}$$

Then the overall processing time used must not exceed the total available time at its assembly line or manufacturing facility:

$$\sum_{p\in M_i} te_{ip}^g Q_{ip}^g - f_i^g W_i^g \le 0 \quad (\forall i\in S,\ \forall g\in G) \quad \text{(A6)}$$

There is usually a minimum amount of the aggregate capacity of a sub-contractor that should be consumed to justify the establishment of a contract. This consideration leads to constraints (A7) where the first term is the total time used at the assembly line or manufacturing facility of subcontractor i in order to manufacture all the products. The second term of the left hand side of the inequality is the minimum time to be used:

$$\sum_{p\in M_i}\sum_{g\in G} te_{ip}^g Q_{ip}^g - \rho_i\sum_{g\in G} f_i^g W_i^g \ge 0, \quad \forall i\in S \quad \text{(A7)}$$

To make a deal with a supplier, the minimum capacity can also be considered. Here, the minimum capacity to be used is a percentage of the total weight of all maximum quantities of raw materials that can be supplied by the supplier:

$$\sum_{p\in R_i} X_{ip} - \left(\rho_i\sum_{p\in R_i} b_{ip}\right)A_i \ge 0 \quad (\forall i\in V) \quad \text{(A8)}$$

The constraints of flow out of suppliers' nodes are given by the equalities below:

$$X_{ip} - \sum_{j\in SP_p^s\cup D}\sum_{k\in K} F_{ijp}^k = 0 \quad (\forall p\in P,\ \forall i\in V_p) \quad \text{(A9)}$$

The constraints of flow out of sub-contractors' nodes are given by the equalities below:

$$\sum_{g\in G} Q_{ip}^g - \sum_{j\in SP_p^s\cup D}\sum_{k\in K} F_{ijp}^k = 0 \quad (\forall p\in P,\ \forall i\in S_p) \quad \text{(A10)}$$

For each product, the quantity that arrives to a node must equal the quantity needed to manufacture next higher assemblies:

$$\sum_{j\in S_p}\sum_{k\in K} F_{jip}^k - \sum_{p'\in P_p^s}\sum_{g\in G}\theta_{pp'}Q_{ip'}^g = 0 \quad (\forall p\in M,\ \forall i\in SP_p^s) \quad \text{(A11)}$$

The quantity of finished products shipped from all its sub-contractors to the demand node must equal the demand of that product:

$$\sum_{i\in S_p}\sum_{k\in K} F_{idp}^k = d_{pd} \quad (\forall p\in C,\ \forall d\in D) \quad \text{(A12)}$$

For each couple of nodes, there is a maximum number of transportation modes that can be used:

$$\sum_{k\in K} Z_{ij}^k \le \tau_{ij} \quad (\forall i\in V\cup S,\ \forall j\in S\cup D) \quad \text{(A13)}$$

The quantity of products shipped between two nodes is limited by the capacity of transportation mode and the number of shipments. While the first set of constraints (A14) expresses the volume capacity and the second set (A15) expresses the weight capacity:

$$\sum_{p\in R_i\cup M_i}\delta_p F_{ijp}^k - \kappa^k U_{ij}^k \le 0 \quad (\forall i\in V\cup S,\ \forall j\in S\cup D,\ \forall k\in K) \quad \text{(A14)}$$

$$\sum_{p\in R_i\cup M_i}\pi_p F_{ijp}^k - \psi^k U_{ij}^k \le 0 \quad (\forall i\in V\cup S,\ \forall j\in S\cup D,\ \forall k\in K) \quad \text{(A15)}$$

The following are logical constraints. The number of shipments between two nodes for a given transportation mode is not nil only if the transportation mode is actually used. This yields to the following constraints:

$$U_{ij}^k - MZ_{ij}^k \le 0 \quad (\forall i\in V\cup S,\ \forall j\in S\cup D,\ \forall k\in K), \quad \text{(A16)}$$

where M is a big number

A site is selected if it is open for one product at least:

$$Y_{ip} - A_i \leq 0 \quad (\forall i \in S \cup V,\ \forall p \in M_i \cup R_i) \tag{A17}$$

The following are constraints on decision variables. The transport variables, the quantities supplied and manufactured by sites are non negative:

$$F^k_{iip} \geq 0 \quad (\forall p \in R \cup M,\ \forall i \in V_p \cup S_p,\ \forall j \in SP^s_p \cup D,\ \forall k \in K) \tag{A18}$$

$$X_{ip} \geq 0 \quad (\forall (p,i) \in R \times V_p \cup M \times S_p) \tag{A19}$$

$$Q^g_{ip} \geq 0 \quad (\forall p \in M\ \forall i \in S_p\ \forall g \in G) \tag{A20}$$

Binary variables:

$$Y_{ip} \in \{0,1\}, \quad \forall (p,i) \in R \times V_p \cup M \times S_p \tag{A21}$$

$$A_i \in \{0,1\}, \quad \forall i \in S \cup V \tag{A22}$$

$$Y_{ip} \in \{0,1\}, \quad \forall p \in R \cup M\ \forall i \in S \cup V \tag{A23}$$

$$W^g_i \in \{0,1\}, \quad \forall i \in S\ \forall g \in G \tag{A24}$$

$$Z^k_{ij} \in \{0,1\} \quad (\forall i \in V \cup S,\ \forall j \in S \cup D,\ \ k \in K) \tag{A25}$$

The number of shipments must be integer:

$$U^k_{ij} \text{ integer } (\forall p \in P,\ \forall i \in V_p \cup S_p,\ \forall j \in SP^s_p \cup D,\ \forall k \in K) \tag{A26}$$

In the case of minimising F_2, the following constraints should be added to the model. No assignment of raw material to supplier if the raw material is not supplied by this supplier:

$$Y_{ip} - X_{ip} \leq 0 \quad (\forall P \in P,\ \forall i \in V_p) \tag{A27}$$

No assignment of manufactured product to plants if the product is not manufactured in this plant

$$Y_{ip} - \sum_{g \in G} Q^g_{ip} \leq 0 \quad (\forall P \in P,\ \forall i \in V_p) \tag{A28}$$

A technology is acquired only if it used to produce at least one product:

$$W^g_i - \sum_{p \in M_i} Q^g_{ip} \leq 0 \quad (\forall i \in S,\ \forall g \in G) \tag{A29}$$

A site is selected if it is open for one product at least:

$$A_i - \sum_{p \in M_i \cup R_i} Y_{ip} \leq 0 \quad (\forall i \in S \cup V) \tag{A30}$$